

PyCon JP 2026

How Python helped me getting the JLPT N1?

From dattebayo! to Leading an Engineering Team in Tokyo

Jose

**How many of you here today
speak Japanese?**

2016 (My only vocabulary)

俺 "ore"

Translation: "I" (Ultra-masculine)

⚠ USE WITH BOSS AT YOUR OWN RISK

だってばよ "dattebayo"

Translation: "Believe it!" (Naruto)

🌀 NOT A REAL WORD, BELIEVE IT!

こんにちは "konnichiwa"

Translation: "Hello" (Normal word)

✅ SAFE TO USE IN SOCIETY

お兄ちゃん "onichan"

Translation: "Big brother" (Cringe at work)

💀 MAXIMUM SOCIAL AWKWARDNESS

2024 (The top)

2024 年 7 月 7 日に、公益財団法人日本国際教育支援協会及び独立行政法人国際交流基金が実施した日本語能力試験に関し、認定結果及び成績を次のとおり証明します。

This is to certify the result and the scores of Japanese-Language Proficiency Test given on July 7, 2024 jointly administered by Japan Educational Exchanges and Services and the Japan Foundation .

発行日 Date of Issue (y/m/d)	2024/09/03
受験番号 Registration No.	132811007
氏 名 Name	JUANJOSE HAFID TENORIO PENIA
生年月日 Date of Birth (y/m/d)	1994/11/26
受験地 Test Site	日本 Japan
レベル Level	N1
結 果 Result	合格 Passed
認定番号 Certification No.	N1A305265J





**How I did it? And how did
Python helped me?**

Is English Really Enough?

"The 'official' language for IT engineers is English, right?"

Well, yes, but not quite like that here.

The Reality of English-Only Tech Roles

A Highly Concentrated Ecosystem

The market for pure English-speaking engineering roles in Japan is remarkably tight

300–500

Total Companies

Total estimated firms in Japan accommodating English-only tech talent.

150–250

Active Startups

Startups actively hiring, heavily driven by Deep Tech and AI sectors.

Severe Logistical Friction

Despite an eagerness to onboard international talent, relocating engineers from overseas presents immense bureaucratic risks and high upfront costs.

40%

Of Job Postings

Explicitly stipulate that the applicant must already be a current resident of Japan.

Not Convinced?

9.1 Work Visas

New JLPT N2 Requirements

Effective **April 15, 2026**, first-time overseas applicants must prove proficiency equivalent to **JLPT N2 (or CEFR B2)**.

This rule targets applicants at **Category 3 and 4 companies** (SMEs and startups) to prevent visa abuse.

While technical roles may be exempt, the **administrative burden of proof is significantly higher**, adding friction to direct sponsorship.

9.2 Permanent Residency

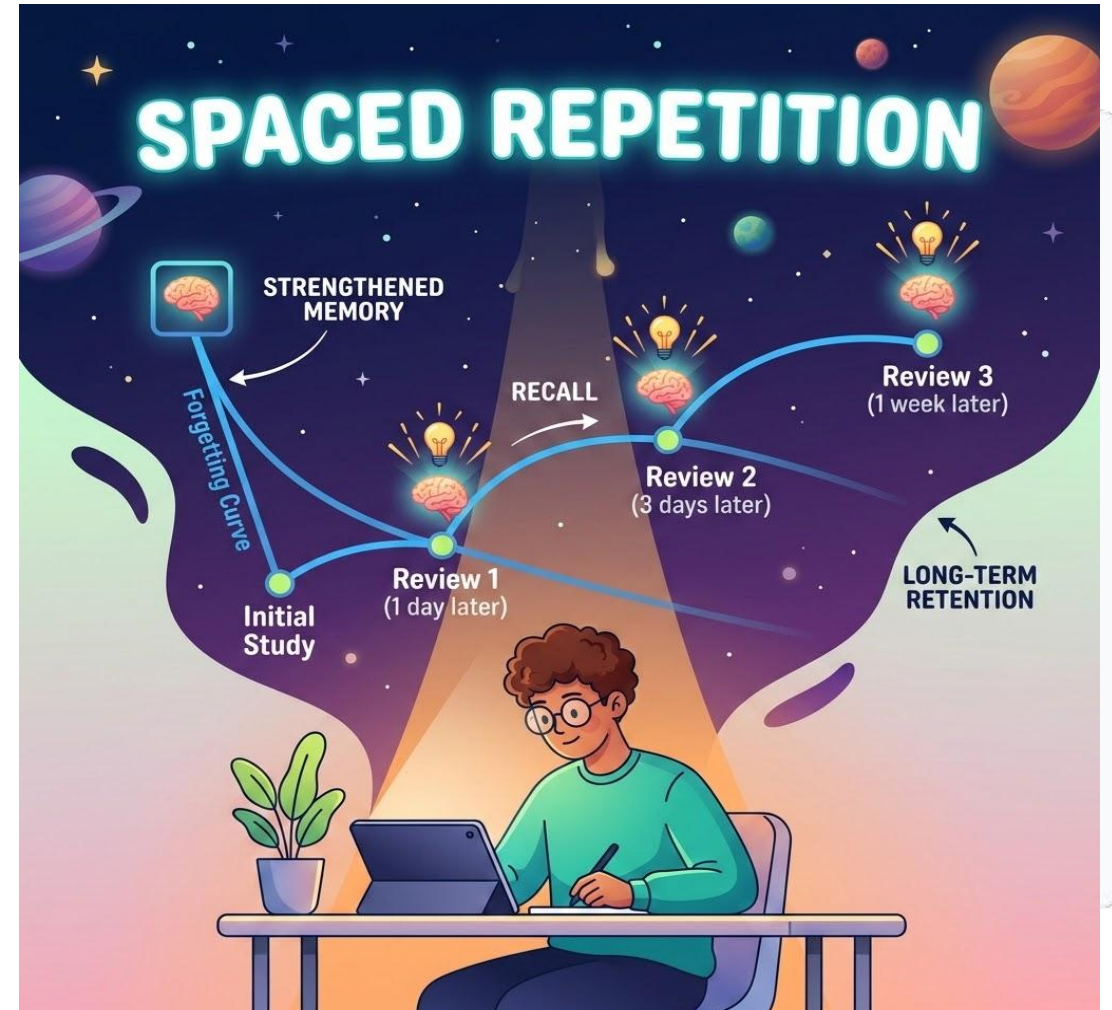
The Push for Mandatory Japanese

A formal **language requirement for PR** is being introduced via a mandatory **"Japanese Language and Lifestyle Learning Programme"**.

With **JLPT N2** as the discussed benchmark, the program will test skills and integration. Guidelines are expected by **FY2027**.

How Do We Learn?

- 🧠 **The Core Mechanic:** Psychologists have proven that reviewing information at dynamically expanding intervals halts the forgetting curve.
- 📅 **The Algorithm:** There's been a decade-long algorithmic competition to find the right interval: SM2 (1987), Free Spaced Repetition Scheduler (2024).
- ∞ **Permafrost Memory:** Some studies show recall spanning 50 years. Human memory forms deep, unshakeable roots under the right conditions, for some reason (?).





**Of course, we have the
most popular app and
influential program for this**







The beginning

JLPT N5 - N4 (2016-2018)

The Beginning: JLPT N5 & N4 (2016–2017)

JLPT N5: The Foundation

2016 • 4 Months Race

The easiest level is mostly focused on **symbol recognition**, including hiragana and katakana.

Features visual identification of around **100 kanji characters** and mastering basic syllabaries using highly-optimized Anki decks.

JLPT N4: Grammar

Jan 2017 – June 2017

Transitioning from isolated symbols to **contextual grammar understanding** and structured syntax.

Years of passive anime exposure transitioned from simple noise to a structured database of recognizable concepts.

JLPT N3: The Engineering problem

June 2017 – December 2017 The Context Deficit Challenge

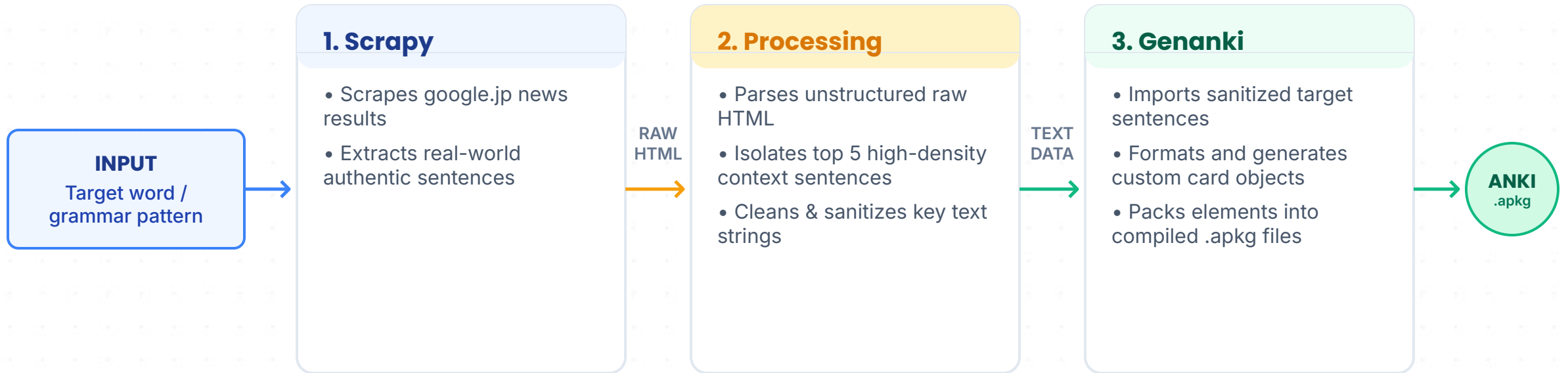
At the time, I chased N3 believing it was a stepping stone to employment. However, the leap in volume was immense—equivalent to **N5 and N4 combined** (500 characters & 1000 words).

Separated from Japan, I lacked the natural environment to apply this vocabulary. **How do you scale context when you have no one to speak to?**

Enter Python



JLPT N3: Adding context





And I stopped

2018 – 2019

2018 – The Gap Year & Pivot

I spent the year working and preparing a high-stakes MEXT scholarship application.

While the MEXT scholarship application ultimately failed, my previous tutor connected me directly with a **Japanese CEO**. I secured an interview—this was the defining moment when intermediate conversational N3 fluency paid off, allowing me to speak and make myself understood.



Japan

My first job (Sep 2019)

Not good enough

N3 vocabulary was good enough for communication, but not for a fully Japanese environment.

I figured I had to get the N2 to get even more words to my brain.

JLPT N2 (Jan 2020 – Dec 2020)

Immersion for N2 (Jan 2020–Dec 2020)

N2 brings **long lectures, specialized grammar**, 1500 words and 1000 Kanjis.

- My old pipeline was working fine for kanjis and grammar but still it was hard to maintain. so I tried the community supported add on **Yomichan**. It would generate Anki flashcards for any content I would see on the web.
- Being immersed tremendously helped especially on the reading and listening sections.
- In the professional aspect, I would learn a lot of **technical stuff..**



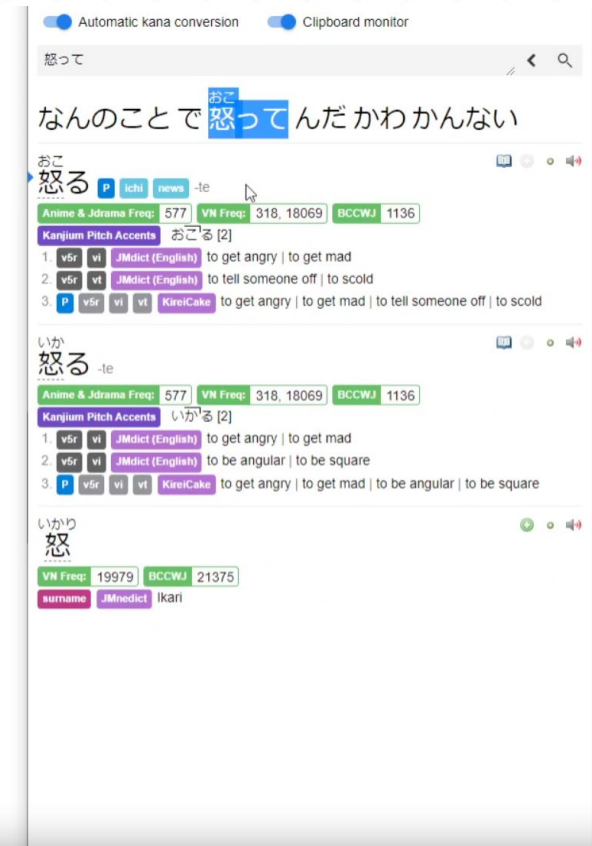
And I stopped

2021 – 2023 (The community moves forward)

Manga-OCR & Transformers

The Python community introduced [manga-ocr](#). Built on a Vision Encoder-Decoder transformer framework, it accurately parses vertical text, furigana, and stylized fonts from images, seamlessly linking native reading to flashcard creation.

The dataset was provided from a compilation of free mangas by the University of Tokyo (manga-109 dataset)



And chappy arrived



ChatGPT

Conquering JLPT N1 (Jan 2024 – July 2024)

Vocabulary & Kanji

Anki Was the Boss

Systematically brute-forced and memorized all **2,136 Joyo Kanji**. Constant spaced repetition formed the core structural foundation of the vocabulary stack.

Grammar Synthesis

ChatGPT Integration

Used as the primary sentence generator. Since the Model Context Protocol (MCP) was only in its early conception, the pipeline relied on direct trust in ChatGPT's raw linguistic outputs.

I would run a script with genanki to generate the flashcards.

Comprehension

Organic Reading & Listening

Allowed comprehension to develop naturally. Trusted the permafrost and used physical books as baseline indicators to measure visual reading speed and pacing over time.

December 2024 (The end)

2024 年 7 月 7 日に、公益財団法人日本国際教育支援協会及び独立行政法人国際交流基金が実施した日本語能力試験に関し、認定結果及び成績を次のとおり証明します。

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So it was over...

A green cartoon owl with large, light-colored eyes and an orange beak. The owl is centered in the frame against a dark background. The text "Not really" is written in white, bold, sans-serif font across the middle of the owl's face.

Not really

The illusion of mastery

The Recall Deficit

I possessed great recognition for kanji characters when reading, but froze when forced to produce them manually.

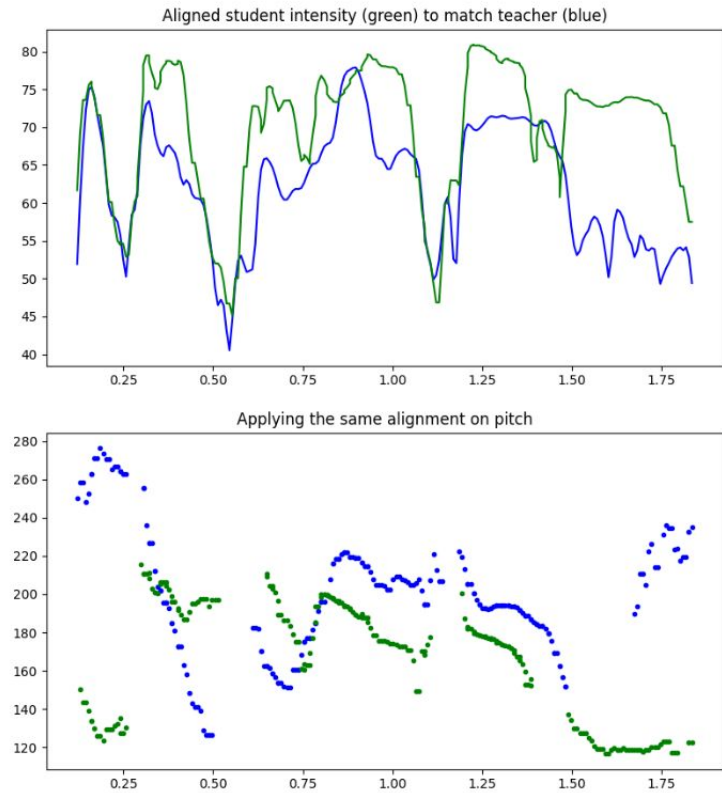
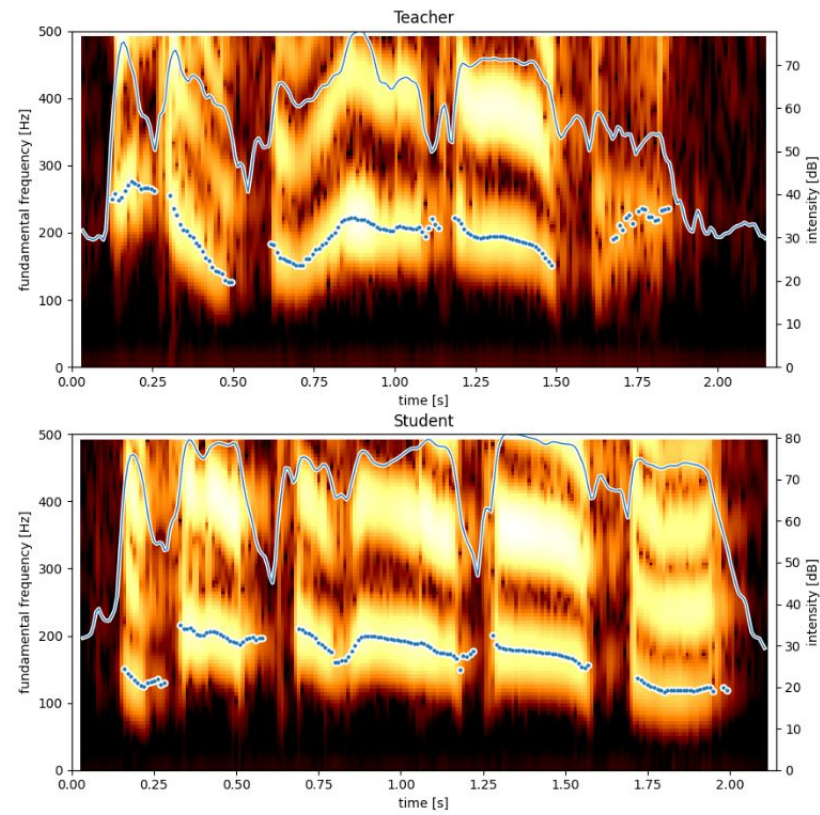
The Pronunciation Blindspot

Despite continuous speaking, I remained entirely oblivious to the fundamental importance of pitch accents.

My partner would constantly point out at my accent mistakes.

**Do you need a
Japanese waifu to
pronounce better?**

Onsei



The Pronunciation gap

Not really

Onsei

An ongoing project. It records your voice, analyzes the pitch contour, and grades your cadence against the standard Tokyo dialect.

NHK Pronunciation Dictionary

The gold standard. While less automated, manually verifying pitch accents against the official NHK database provides 100% reliable native audio.

The 2025/2026 Learning Stack



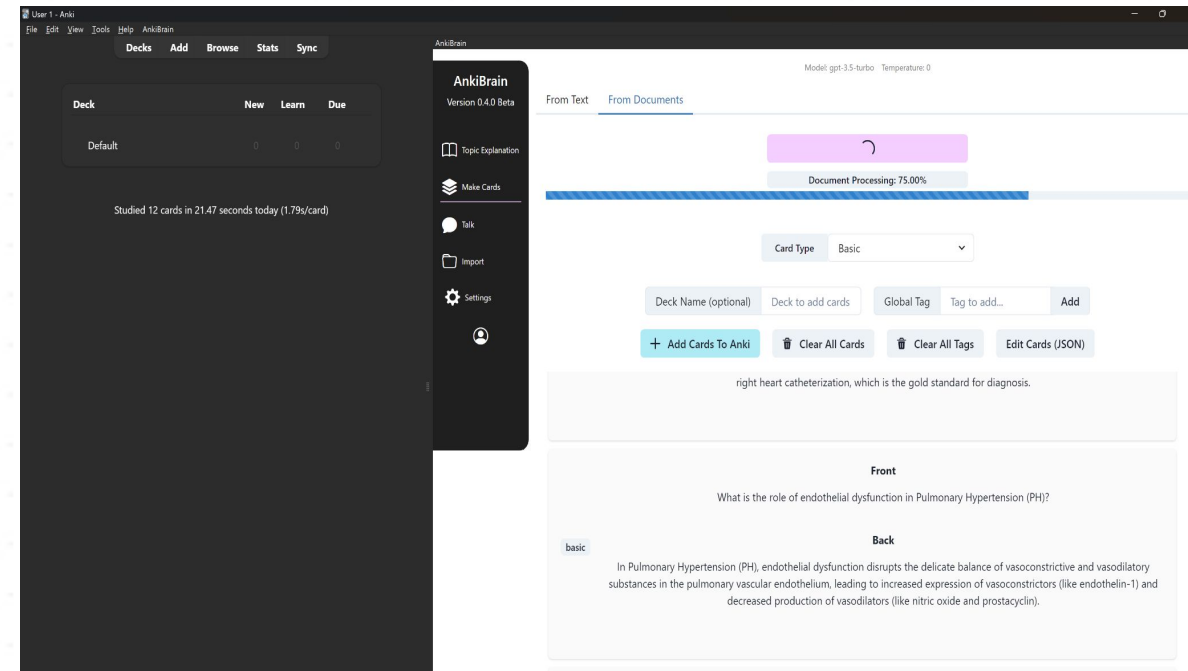
Claude Code & Genanki: Instead of simple scrapers, use agentic LLMs to parse specific datasets (like industry buzzwords) and automatically construct `.apkg` files.



Anki Brain: Advanced integrations now allow you to feed entire PDFs, technical ePubs, or corporate documents into an AI, extracting the vocabulary instantly.



Organic Returns: Eventually, the goal is to step away from the flashcards and let the language become second nature through pure consumption.



A never-ending journey

- ✓ **Organic is King:** Aim for organic learning. It's more fun that way.
- 👂 **Recognition vs Recall. :** Living in the country makes listening second nature but it's trickier to learn pronunciation. You can recognize kanjis but cannot write them (even the Japanese cannot do it sometimes).
- 📈 **The Professional Multiplier:** Having reliable, business-level Japanese is a massive career booster: broad interview access, credibility and negotiation edge.



**Will this matter in
50 years?**

I believe so. Language is your ultimate insurance.

Questions?

Thank you for listening!

<https://github.com/jota12x/pycon-jp-2026-docs>

